MERCHANT SALES-BASED FINANCING STRATEGY SIMULATOR

MSBF Module 1/2 v2.1 Technical Successor

Technical Validation and Evidence Report

Report date	2026-09-04
Report version	R1
Validation classification	Publication-candidate support; not publication authority

CONTROLLED STATUS: Technical successor accepted; version-separated integration authorized; public supersession and publication not authorized.

Deterministic synthetic data | Non-production | No real funds movement

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1. Validation objective

This report documents how the change log and data reconciliation were independently re-performed from physical artifacts. It is designed so an auditor or technical reviewer can distinguish:

- source facts from interpretive statements;
- direct before/after comparisons from taxonomy-related comparisons;
- accepted predecessor authority from successor candidate evidence;
- business-output changes from identity-only changes;
- final PASS evidence from failed historical provenance.

2. Physical sources used

Source family	Representative artifact	Use
Prior public code/repository	Public v2.0.0 Git HEAD and accepted stage sign-offs	Comparator authority
Prior data	M2.2 summary; M2.4 and M2.5 formal sign-offs	Prior counts and values
R4 strategy	R1B Phase 2C R4 final-decision, candidate, gate, proof and comparison files	Decision logic and migration
R6 activation	M2.4 R6 closure receipt and 130 advance rows	Funded counts/amounts
R17 Wave A	Final M2.5-M2.10 exports, validation and transformed sources	Portable lifecycle predecessor
R26 R2	Failed runtime receipt and transformed sources	Policy implementation and FAIL provenance
R26 resume	Wave A exports, M2.11 populations, G3-v2 controls and identities	Accepted technical successor
R23 repository audit	Repository patch, diff index, candidate tree and audit decision	Pre-R26 integration evidence
Power BI evidence	P01-P04 physical evidence and QA	Presentation/consumption state

All raw sources remain outside this public documentation package. This package includes derived registers, reports, exact source names and identities, but not private database evidence or local Desktop artifacts.

3. Re-performance controls

Control ID	Area	Observed	Status
PRIOR_PUBLIC_REPOSITORY	Public v2.0.0 HEAD reconstruction	1,934 files / public validator PASS	PASS
PRIOR_DATA_SNAPSHOT	M2/G3 CSV snapshot	253 files / 1,115,469 rows / 6,954 columns	PASS
R4_DECISION	R4 strategy outputs	1,500 final decisions / 6,519 candidates / 84,228 gate rows	PASS
R4_MIGRATION	Predecessor-successor migration	1,500 rows; row/column totals reconcile	PASS
R6_ACTIVATION	M2.4 activation	130 accounts; 120/120 positive; 20/20 negative	PASS
R17_WAVE_A	M2.5-M2.10 rebuild	115/115 controls	PASS
R17_ROUNDING	Known retry recurrence	0.00 reconciliation and exposure variance	PASS
R26_POLICY_STATIC	Monitoring-policy static re-performance	16/16 controls	PASS
R26_MONITORING	Daily monitoring checkpoints	18/18 checkpoints; 15,600 rows	PASS
R26_DOWNSTREAM	M2.6-M2.10 R17/R26 business-output comparison	5/5 stage exports exact	PASS
R26_RESUME	Retained-final resume	Wave A re-execution = false; nine exports exact	PASS_WITH_OBSERVATIONS
R26_M211	M2.11 simulation	8 / 52,152 / 12,000 / 1,040	PASS
R26_G3V2	G3-v2 certification	70/70	PASS
R23_REPOSITORY	Pre-R26 repository candidate	508 paths; zero accepted-stage changes	PASS_PRE_R26
P01_P04	Power BI successor public-scope pages	P01-P04 complete and tested; P03 2/0/1,500	PASS_CANDIDATE
P05_P06	Strategy-profile public showcase	Held from publication	HOLD
FINAL_R26_REPOSITORY	Exact final v2.1 Git integration	Not yet built/frozen	PENDING

4. Decision reconciliation method

The migration matrix is generated by a crosstab of the exact 1,500-row `predecessor_outcome_code` and `successor_governed_outcome_code` fields.

Required invariants:

```
Source rows                                1,500
Migration-matrix grand total               1,500
Predecessor row totals match source counts  PASS
Successor column totals match R4 outcomes   PASS
Missing/extra application-scenario keys     0
Duplicate comparison keys                   0
```

The public register is `registers/03_DECISION_OUTCOME_MIGRATION_MATRIX.csv`.

5. M2.4 funding reconciliation method

Funded paths are counted from the physical `synthetic_advance_funding` rows. Funded amount is the direct sum of `funded_amount`; scenario splits are filtered by `scenario_code`.

Required invariants:

```
Advance rows                            130
Unique application-scenario keys        130
Unique synthetic accounts               130
Unique synthetic advances               130
Baseline rows                            103
Recession Energy rows                     27
Total funded amount             $1,729,500
Real-funds movement flags true              0
```

6. Monitoring reconciliation method

R17 and R26 monitoring facts are sorted and joined by:

```
module1_run_id
+ scenario_code
+ merchant_application_id
+ synthetic_advance_id
+ monitoring_day_index
+ monitoring_date
```

Every shared field is compared across all 15,600 rows. Numeric values are compared at stable decimal precision; null-state equality is separately required. The complete result is `registers/05_R17_TO_R26_MONITORING_FIELD_RECONCILIATION.csv`.

Field	Different	Unchanged	Result
cumulative_pace_ratio	9249	6351	BUSINESS_CALIBRATION_CHANGE
cumulative_shortfall_amount	4840	10760	BUSINESS_CALIBRATION_CHANGE
monitoring_status_code	3300	12300	BUSINESS_CALIBRATION_CHANGE
cumulative_pace_watch_alert_flag	1656	13944	BUSINESS_CALIBRATION_CHANGE
daily_shortfall_alert_flag	1084	14516	BUSINESS_CALIBRATION_CHANGE
cumulative_pace_high_alert_flag	591	15009	BUSINESS_CALIBRATION_CHANGE
stress_status_floor_applied_flag	216	15384	BUSINESS_CALIBRATION_CHANGE
actual_remittance_amount	0	15600	EXACT_NONREGRESSION
cumulative_remittance_amount	0	15600	EXACT_NONREGRESSION
horizon_overrun_alert_flag	0	15600	EXACT_NONREGRESSION
liquidity_stress_alert_flag	0	15600	EXACT_NONREGRESSION
merchant_application_id	0	15600	EXACT_NONREGRESSION
module1_run_id	0	15600	EXACT_NONREGRESSION
monitoring_date	0	15600	EXACT_NONREGRESSION
monitoring_day_index	0	15600	EXACT_NONREGRESSION
paid_off_flag	0	15600	EXACT_NONREGRESSION
principal_exposure_proxy	0	15600	EXACT_NONREGRESSION
receivable_balance_after	0	15600	EXACT_NONREGRESSION
scenario_code	0	15600	EXACT_NONREGRESSION
source_eligible_pos_sales	0	15600	EXACT_NONREGRESSION
synthetic_advance_id	0	15600	EXACT_NONREGRESSION
zero_sales_streak_alert_flag	0	15600	EXACT_NONREGRESSION

The 11 R26-only columns are validated as an additive alert-family/policy layer rather than treated as missing R17 fields.

7. Downstream nonregression method

M2.6-M2.10 R17 and R26 exports are sorted on their stable application/account keys. Every shared business field is compared. Results:

Stage	Rows	Shared columns	Different cells	Result
M2.6	130	13	0	EXACT
M2.7	130	18	0	EXACT
M2.8	130	23	0	EXACT
M2.9	130	22	0	EXACT
M2.10	130	19	0	EXACT

This proves the R26 policy calibration did not silently alter final intervention, setup, servicing execution, account certification, or portfolio performance outputs.

8. Strategy-profile interpretation method

Application simulations are grouped by application-scenario key across the eight profiles. The validator counts distinct strategy outcomes, selected candidates, and selected amounts.

Test	Result
Application-scenario paths	1,500
Paths with profile outcome migration	0
Paths with selected-candidate change	249
Paths with selected-amount change	247
Candidate-bearing paths	380
Candidate-change rate where applicable	65.5%

The conclusion that P05/P06 should be held is therefore a business-interpretation decision grounded in physical data, not a technical failure of the simulation.

9. Identity and provenance validation

Identity	Scope	Value	Status
Prior public Git HEAD	v2.0.0 public predecessor	46fb1f9198475678edc9cabdeefd2f91a59e49c2	CONTROLLING_PUBLIC
Prior Module 2 release commit	v2.0.0 peeled tag	500ec776ca25f1d4ade4eb45ecddb8df5e4f2e65	HISTORICAL_ACCEPTED
R23 repository candidate HEAD	v2.1.0-rc2 pre-R26 candidate	58b4dbda229d7176431cb09eecd1e3e3ec1feeee	ACCEPTED_PRE_R26_BASELINE
R26 result ZIP	Accepted technical successor	113450c7167c8a5e1b40497c4e39506d276aa0e057bbeb3f316561af5d4096c3	ACCEPTED_TECHNICAL_SUCCESSOR
R26 resume database	Accepted execution evidence	msbf_r1_r26_resume_g3v2_20260902_070655	ACCEPTED_TECHNICAL_SUCCESSOR
R26 monitoring policy	Owner-approved calibration	6c6a6f8103bb7276c5d1063d900d96070903e82b5f7143431bc150556e092027	ACCEPTED
R26 source root	M2.11 semantic source identity	618ac7ad9278ad0c3876601eb75336bb166132d412f9eae221a145cda770d41f	ACCEPTED
R26 simulation bundle	M2.11 physical simulations	569ea013b573b9c1e49592cbca0f078861b8d11f55462c3c49d718462c5b12ee	ACCEPTED
R26 control root	G3-v2 controls	3d9e6f264bd220e7a80a7e433945d37a41ade548581d273218b0e3df8b6f2498	ACCEPTED
R26 Power BI source bundle	Successor source carrier	b858965c48cd3aa0ca8ef9277305c2b2377db11cec03c1f48690c772764f98af	ACCEPTED_SOURCE_IDENTITY
R26 report root	G3-v2 master/detail reports	c053dc064ab32a4d21864f943d7efb1d70adffc31701bddfbf26daf2db1b802d	ACCEPTED
R26 certification	G3-v2 certification ID	G3V2_R26_C053DC064AB32A4D21864F94	ACCEPTED
R26 R2 failed result	Historical failed provenance	f140d5f51d464b379dc218902f1b61844914a86207665b4ff0699aab8bbe4329	FAIL_HISTORICAL_PROVENANCE

Critical provenance rule:

- `f140d5...4329` remains the failed R26 R2 result.
- `113450...096c3` is the accepted resume result.
- The failed result is not relabeled or overwritten.
- R6 and the retained R2 final database both pass nonmutation.

10. Repository validation

R23 independently establishes the version-separated repository pattern, but not the final R26 tree.

Control	Observed
Accepted base HEAD	46fb1f9198475678edc9cabdeefd2f91a59e49c2
R23 candidate HEAD	58b4dbda229d7176431cb09eecd1e3e3ec1feeee
Changed paths	508
Added / modified	501 / 7
Accepted stage directory changes	0
Accepted authority-file hash changes	0
Binary patch apply and candidate reproduction	PASS
Candidate overlay validator	PASS
Public release validator	PASS
Final R26 integrated repository	PENDING

The final v2.1 file-level validation must be rerun after R26 integration. This report does not substitute the R23 pre-R26 file list for that future evidence.

11. Package-level validation requirements

The companion validator verifies:

- required reports/registers/figures exist;
- CSVs parse and required columns exist;
- all key population formulas reconcile;
- decision migration totals equal 1,500;
- M2.4 path counts and funded amounts reconcile;
- R17/R26 monitoring keys and row counts match;
- M2.6-M2.10 cell comparisons are exact;
- M2.11 populations equal 8 / 52,152 / 12,000 / 1,040;
- G3-v2 controls equal 70/70 PASS;
- report status wording does not authorize publication;
- no local absolute paths or temporary sandbox links appear in public files;
- manifest sizes and SHA-256 values match;
- deterministic package replay is possible.

12. Validation disposition

The evidence supports the following conclusion:

```
Technical change specification        PASS
Data reconciliation                   PASS
Code-logic change register            PASS
R26 identity reconciliation           PASS
Predecessor nonregression              PASS
R23 pre-R26 repository validation      PASS
Final R26 repository integration       PENDING
Public supersession/publication        NOT AUTHORIZED
```

The package is suitable for inclusion in the future combined integration-delta audit. It is not by itself a release approval.